Building the GuaranIA LLM

This is a living document that describes the approach followed to develop the LLMs for the project GuaranIA. The building process follows the BYOL methodology, which is implemented through the in-house framework Guaratune. **The main idea is to adapt a multilingual open-source base model (e.g., Goolge Gemma-4) to Guarani while aiming to retain its existing capabilities.**

BYOL methodology in a nutshell

The BYOL methodology consists of three steps: `continual pretraining (CPT)`, where the base model continues training on a curated langage corpus, `supervised fine-tuning (SFT)`, where the CPT model is futher trained on instruction-and-response examples, and `model merging`, where the adapted model is combined with the original model in weight space. See the BYOL paper for more information about the framework.

GuaranIA approach

Continual pretraining (CPT)

Finding the dataset configuration

The guarani corpus Kuatia augmented with Spanish and English FineWeb-Edu samples are proposed to be used to build the GuaranIA LLMs. The three sources were combined into eigth different data configurations for `full CPT` (i.e., all model weights are updated) of Gemma-4 E4B-PT to identify the most promising configuration.

- `C1`: Kuatia (`43,677,319` tokens)
- `C2`: Kuatia without synthetic documents (`27,994,685` tokens)
- `C3`: Kuatia + 20% Spanish FineWeb-Edu (`52,415,631` tokens)
- `C4`: Kuatia without synthetic documents + 20% Spanish FineWeb-Edu (`36,732,997` tokens)
- `C5`: Kuatia + 20% FineWeb-Edu (`52,412,827` tokens)
- `C6`: Kuatia without synthetic documents + 20% FineWeb-Edu (`36,730,193` tokens)
- `C7`: Kuatia + 10% Spanish FineWeb-Edu + 10% FineWeb-Edu (`52,415,521` tokens)
- `C8`: Kuatia without synthetic documents + 10% Spanish FineWeb-Edu + 10% FineWeb-Edu (`36,732,887` tokens)

CPT Hyperparameters

For Gemma-4 E4B CPT, the experiment configuration uses a sequence length of `2,048`, four epochs, a per-device batch size of `1`, and a learning rate of `2e-5` with a cosine scheduler and `2e-6` minimum. Training uses packed sequences, `bfloat16`, AdamW (betas `0.9` and `0.999`), `3%` warmup, and gradient clipping at `1.0`. Evaluation runs at the end of each epoch. Full CPT uses gradient accumulation of `32` and weight decay of `0.01`.

Evaluation

The eight trained variants (C1-C8) are evaluated against the base model on Guarani, Spanish, and English benchmarks, including language understanding, reading comprenhension, question answering, translation, and perplexity tasks. The table below lists each benchmark, its language (`GN` = Guarani, `ES` = Spanish, `EN` = English), the capability and task it assesses, as well as the metric, number of evaluation instances, number of shots, and score split.

Benchmark	Language	Capacity	Task	Metric	Instances	Shots	Score Split
ARC Challenge	ES	Specialized reasoning (science)	Multiple-choice QA	Normalized accuracy	299	25	validation
ARC Challenge	EN	Specialized reasoning (science)	Multiple-choice QA	Normalized accuracy	299	25	validation
ARC Easy	ES	Specialized reasoning (science)	Multiple-choice QA	Normalized accuracy	570	0	validation
ARC Easy	EN	Specialized reasoning (science)	Multiple-choice QA	Normalized accuracy	570	0	validation
Belebele	GN	Reading comprehension	Multiple-choice QA	Accuracy	900	1	test
Belebele	ES	Reading comprehension	Multiple-choice QA	Accuracy	900	1	test
Belebele	EN	Reading comprehension	Multiple-choice QA	Accuracy	900	1	test
COPA	ES	Causal reasoning	Multiple-choice QA	Accuracy	100	0	validation
COPA	EN	Causal reasoning	Multiple-choice QA	Accuracy	100	0	validation
FLORES+	GN	Translation	Generation	BLEU, chrF++	997	1	dev
FLORES+	ES	Translation	Generation	BLEU, chrF++	997	1	dev
FLORES+	EN	Translation	Generation	BLEU, chrF++	997	1	dev
Global MMLU Lite	GN	General knowledge	Multiple-choice QA	Accuracy	400	5	test
Global MMLU Lite	ES	General knowledge	Multiple-choice QA	Accuracy	400	5	test
Global MMLU Lite	EN	General knowledge	Multiple-choice QA	Accuracy	400	5	test
Global PIQA	ES	Common sense & inference	Multiple-choice QA	Normalized accuracy	100	0	test
Global PIQA	EN	Common sense & inference	Multiple-choice QA	Normalized accuracy	100	0	test
GPQA Main	ES	Specialized reasoning (biology, physics, chemistry)	Multiple-choice QA	Accuracy	448	5	train
GPQA Main	EN	Specialized reasoning (biology, physics, chemistry)	Multiple-choice QA	Accuracy	448	5	train
HellaSwag	ES	Common sense & inference	Multiple-choice QA	Normalized accuracy	9,374	10	val
HellaSwag	EN	Common sense & inference	Multiple-choice QA	Normalized accuracy	10,042	10	validation
MGSM	GN	Specialized reasoning (mathematics)	Generation	Exact match	142	5	test

Benchmark	Language	Capacity	Task	Metric	Instances	Shots	Score Split
MGSM	ES	Specialized reasoning (mathematics)	Generation	Exact match	250	0	test
MGSM	EN	Specialized reasoning (mathematics)	Generation	Exact match	250	0	test
MultiWikiQA	GN	Reading comprehension	Generation	Exact match, F1	5,003	2	train
MultiWikiQA	ES	Reading comprehension	Generation	Exact match, F1	5,001	2	train
MultiWikiQA	EN	Reading comprehension	Generation	Exact match, F1	4,999	2	train
TruthfulQA MC1	ES	Truthfulness and factuality	Multiple-choice QA	Accuracy	789	6	val
TruthfulQA MC1	EN	Truthfulness and factuality	Multiple-choice QA	Accuracy	817	6	validation
WNLI	GN	Common sense & inference	Multiple-choice QA	Accuracy	49	5	dev
WNLI	ES	Common sense & inference	Multiple-choice QA	Accuracy	71	5	validation
WNLI	EN	Common sense & inference	Multiple-choice QA	Accuracy	71	5	validation
XStoryCloze	ES	Common sense & inference	Multiple-choice QA	Accuracy	1,511	5	eval
XStoryCloze	EN	Common sense & inference	Multiple-choice QA	Accuracy	1,511	5	eval

> [!Important] Complementing the mentioned benchkmarks, we assess the language modeling capacity of the adapted model through `perplexity` on COREGUAPA, a high-quality proprietary corpus curated by the Paraguayan Secretariat of Linguistic Policy.

> [!Note] A Guarani version of `Global MMLU Lite`, `MGSM`, and `WLNI` were created by language experts and are available at the framework Guaratune repo.

Data configuration results

Perplexity

Across the **`COREGUAPA` perplexity** evaluation, every `CPT` variant scores below the `Gemma-4 E4B` base model on all three reported metrics (i.e., `word perplexity`, `byte perplexity`, and `bits per byte`). As shown in the table below, **`C7`** (the data configuration that includes `Kuatia` both real and synthetic plus English and Spanish FineWeb-Edu) **has the lowest `word perplexity`, `byte perplexity`, and `bits per byte` values**, compared with the base model.

Perplexity metric	base	C1	C2	C3	C4	C5	C6	C7	C8
word perplexity	3058.27	469.16	533.09	476.78	542.04	485.26	553.65	**463.10**	521.08
byte perplexity	2.64	2.11	2.14	2.11	2.14	2.11	2.15	**2.10**	2.13
bits per byte	1.40	1.07	1.10	1.08	1.10	1.08	1.10	**1.07**	1.09

Note: In bold the lowest value per row

The other `CPT` variants are close on `byte perplexity` (`2.11`–`2.15`) and `bits per byte` (`1.07`–`1.10`), so `C7`'s lead on those measures is small. **Lower values indicate better prediction of the evaluation text**. The report covers `66` `COREGUAPA` examples, so these results describe performance on thist high-quality set but they do not alone represent broad language ability, which is checked next.

Guarani performance

The full CPT variants show a mixed pattern across the Guarani benchmarks. **All eight configurations improve on the base model in each of the four `FLORES+` translation directions**. `C3` has the highest `chrF++` for `English→Guarani` (`30.82` vs. `8.48` for the base), and `C1` is highest for `Spanish→Guarani` (`29.81` vs. `9.30`). `C7` produces the best scores for both `Guarani→English` (`44.47` vs. `37.94`) and `Guarani→Spanish` (`37.11` vs. `33.40`). Overall, **`C7` has the strongest translation profile**, leading both directions from Guarani and performing close to the top variants on translations into Guarani, as shown in the table below.

benchmark	base	C1	Gain C1	C2	Gain C2	C3	Gain C3	C4	Gain C4	C5	Gain C5	C6	Gain C6	C7	Gain C7	C8	Gain C8
FLORES+ EN->GN (chrF++)	8.48	30.24	+21.76	29.56	+21.08	**30.82**	+22.34	30.24	+21.76	30.45	+21.97	30.02	+21.54	30.05	+21.57	29.88	+21.40
FLORES+ ES->GN (chrF++)	9.30	**29.81**	+20.51	28.45	+19.15	29.69	+20.39	29.53	+20.23	29.66	+20.36	29.49	+20.19	29.60	+20.30	29.29	+19.99
FLORES+ GN->EN (chrF++)	37.94	43.96	+6.02	43.35	+5.41	43.85	+5.91	43.00	+5.06	44.16	+6.22	42.67	+4.73	**44.47**	+6.53	43.18	+5.24
FLORES+ GN->ES (chrF++)	33.40	36.72	+3.32	36.37	+2.97	36.94	+3.54	36.52	+3.12	36.59	+3.19	35.68	+2.28	**37.11**	+3.71	36.27	+2.87

Note: In bold the highest value per row. Gain columns show signed score-point differences from the corresponding base model.

On the other tasks, gains are limited: **`C7` has the highest `Belebele` accuracy** (`55.56` vs. `52.22` for the base), and `C8` has the highest `WNLI` accuracy (`59.18` vs. `44.90`). As the table below outlines, every CPT variant scores below the base on Guarani `Global MMLU Lite` (`46.00`) and `MGSM` exact match (`9.86`), while all variants also trail the base on `MultiWikiQA` F1 (`42.83`). However, **C7 comes close to the base on the first two tasks**, with differences of only `0.50` points on `Global MMLU Lite` and `0.71` points on `MGSM`.

benchmark	base	C1	Gain C1	C2	Gain C2	C3	Gain C3	C4	Gain C4	C5	Gain C5	C6	Gain C6	C7	Gain C7	C8
Belebele GN (acc)	52.22 (48.95, 55.49)	55.44 (52.19, 58.69)	+3.22	50.11 (46.84, 53.38)	-2.11	52.67 (49.41, 55.93)	+0.45	53.00 (49.74, 56.26)	+0.78	52.89 (49.63, 56.15)	+0.67	52.56 (49.30, 55.82)	+0.34	**55.56 (52.31, 58.81)**	+3.34	55.11 (51.86, 58.36)
WNLI GN (acc)	44.90 (30.83, 58.97)	44.90 (30.83, 58.97)	+0.00	44.90 (30.83, 58.97)	+0.00	40.82 (26.92, 54.72)	-4.08	42.86 (28.86, 56.86)	-2.04	44.90 (30.83, 58.97)	+0.00	38.78 (25.00, 52.56)	-6.12	48.98 (34.84, 63.12)	+4.08	**59.18 (45.28, 73.08)**
Global MMLU Lite GN (acc)	**46.00 (41.11, 50.89)**	44.25 (39.38, 49.12)	-1.75	42.50 (37.65, 47.35)	-3.50	43.75 (38.88, 48.62)	-2.25	44.50 (39.62, 49.38)	-1.50	44.50 (39.62, 49.38)	-1.50	43.00 (38.14, 47.86)	-3.00	45.50 (40.61, 50.39)	-0.50	44.75 (39.87, 49.63)
MGSM GN (exact_match)	**9.86 (4.94, 14.78)**	4.23 (0.91, 7.55)	-5.63	8.45 (3.86, 13.04)	-1.41	4.93 (1.36, 8.50)	-4.93	7.75 (3.34, 12.16)	-2.11	7.04 (2.82, 11.26)	-2.82	6.34 (2.32, 10.36)	-3.52	9.15 (4.39, 13.91)	-0.71	6.34 (2.32, 10.36)
MultiWikiQA GN (F1)	**42.83 (41.66, 44.00)**	36.15 (35.05, 37.25)	-6.68	32.13 (31.07, 33.19)	-10.70	36.79 (35.68, 37.90)	-6.04	32.57 (31.51, 33.63)	-10.26	34.44 (33.35, 35.53)	-8.39	34.55 (33.45, 35.65)	-8.28	36.32 (35.21, 37.43)	-6.51	36.25 (35.13, 37.37)

Note: In bold the highest value per row. In parentheses the 95% confidence intervals (CI), which are approximate normal intervals computed as the score or average ± 1.96 times its standard error. Gain columns show signed score-point differences from the corresponding base model.

Considering its leading results on both translation directions from Guarani, strong `Belebele` score, and near-base results on `Global MMLU Lite` and `MGSM`, **`C7` highlights as the most promising variant for the Guarani capabilities emphasized here**, although `C8` showed the highest displayed average across all Guarani tasks (`37.81` vs. `37.42` of `C7`). So `C7` recommendation reflects the balance of capabilities described above rather than the aggregate average alone.

Deep-dive into the guarani mixed performance

The adapted model's mixed performance on the Guarani benchmarks may have three explanations. First, the base model's existing Guarani capabilities may limit the gains CPT can deliver on these tasks. Second, the CPT data may not provide enough additional information to substantially improve the model's task performance. Third, the size of the adapted model is too large compared with the volume of data, producing less promising gains.

To further investigate the first two explainations, we run additional experiments with a previous generation of the Gemma base model (`Gemma-3 4B`), which passes through a `full CPT` process using the `C7` data configuration. The results of these experiments can use to compare the performance of base models and also to look at the performance gains. If the performance of the `Gemma-3` base is noticeable inferior than the performance of the `Gemma-4` base and the gains of the adapted `Gemma-3` are larger than the adapted `Gemma-4` we can confirm the superior capacity of the latter to solve the guarani tasks. On the other hand, if the performance and gains are similar, the CPT data do not add much more benefits to improve the model's performance.

The table below shows that `Gemma-4 E4B` starts from a higher Guarani average than `Gemma-3 4B` (`31.66` vs. `22.37`) and also reaches a higher average after `C7` CPT (`37.42` vs. `33.18`). However, the average gain is larger for `Gemma-3 4B` (`+10.81` vs. `+5.76`). This pattern is consistent with the idea that the stronger base model has less room to improve through CPT, while the lower-scoring base model has more headroom. Compared with `Gemma-4 E4B`, `Gemma-3 4B` gains more from CPT on almost every tasks, except in `WNLI`. In the case of translation we see mixed results. `Gemma-4 E4B` gains more on translation into Guarani, while `Gemma-3 4B` shows larger gains for translation from Guarani. **From these results, it seems that, so far, what might explain the mixed performance of the adapted `Gemma-4` is the superior capacity of its baseline, not the trained data content.**

Benchmark	Gemma-3-Base	Gemma-4-Base	Gemma-3-4B-C7	Gain Gemma-3-C7	Gemma-4-E4B-C7	Gain Gemma-4-C7
Global MMLU Lite GN (acc)	36.00	46.00 (+10 w.r.t. `Gemma-3`)	44.25	+8.25	45.50	-0.50
MGSM GN (exact_match)	4.93	9.86 (+4.93)	5.63	+0.70	9.15	-0.71
WNLI GN (acc)	40.82	44.90 (+4.08)	42.86	+2.04	48.98	+4.08
Belebele GN (acc)	34.11	52.22 (+18,11)	43.33	+9.22	55.56	+3.34
MultiWikiQA GN (F1)	16.93	42.83 (+25,90)	33.74	+16.81	36.32	-6.51
FLORES+ EN->GN (chrF++)	7.93	8.48 (+0,55)	27.21	+19.28	30.05	+21.57
FLORES+ ES->GN (chrF++)	8.14	9.30 (+1,16)	27.54	+19.40	29.60	+20.30
FLORES+ GN->EN (chrF++)	27.05	37.94 (+10,89)	39.62	+12.57	44.47	+6.53
FLORES+ GN->ES (chrF++)	25.45	33.40 (+7,95)	34.45	+9.00	37.11	+3.71
Average	22.37	31.66 (+9,29)	33.18	+10.81	37.42	+5.76

Note: Gain columns show signed score-point differences from the corresponding base model.

To check the third possible explanation of the mixed performance, we conducted CPT on a smaller variant of Gemma-4 (`Gemma-4 E2B`). Results are outlined in the table below.

Benchmark	Gemma-4-E2B-Base	Adapted Gemma-4-E2B	Gain E2B	Gemma-4-E4B-Base	Adapted Gemma-4-E4B	Gain E4B
Global MMLU Lite GN (acc)	36.25	41.00	+4.75	46.00	45.50	-0.50
MGSM GN (exact_match)	2.11	1.41	-0.70	9.86	9.15	-0.71
WNLI GN (acc)	38.78	51.02	+12.24	44.90	48.98	+4.08
Belebele GN (acc)	34.56	37.22	+2.66	52.22	55.56	+3.34
MultiWikiQA GN (F1)	23.36	30.77	+7.41	42.83	36.32	-6.51
FLORES+ EN->GN (chrF++)	8.38	26.34	+17.96	8.48	30.05	+21.57

Benchmark	Gemma-4-E2B-Base	Adapted Gemma-4-E2B	Gain E2B	Gemma-4-E4B-Base	Adapted Gemma-4-E4B	Gain E4B
FLORES+ ES->GN (chrF++)	9.01	26.72	+17.71	9.30	29.60	+20.30
FLORES+ GN->EN (chrF++)	28.18	35.04	+6.86	37.94	44.47	+6.53
FLORES+ GN->ES (chrF++)	26.61	31.85	+5.24	33.40	37.11	+3.71
Average	23.03	31.26	+8.23	31.66	37.42	+5.76

Note: Gain columns show signed score-point differences from the corresponding base model.

The smaller `Gemma-4 E2B` gains more on average from `C7` than `Gemma-4 E4B` (`+8.23` vs. `+5.76` points). `E2B` also gains more on Guarani `Global MMLU Lite`, `WNLI`, `MultiWikiQA`, and translation from Guarani; `E4B` gains more on `Belebele` and translation into Guarani. Neither model improves on `MGSM`. This pattern is consistent with the `C7` corpus providing a better data-to-model-size balance for `E2B`, but it is not conclusive: `E4B` starts from a higher average and still finishes with the higher score (`37.42` vs. `31.26`), while the relative gains vary by task. The results support the model-size hypothesis as one possible explanation for the mixed performance, rather than showing that model size alone causes it.

Multilingual performance

Across Spanish and English benchmarks, the CPT variants sometimes score above the base and sometimes below it, as the tables below depict. Although the average scores are generally lower, the gaps from the base are modest. The smallest `Spanish` gap is `2.83` points (`C8=62.82` vs. `base=65.65`), and the smallest English gap is `2.20` points (`C5=63.15` vs. `base=65.35`). This overall proximity suggests that **multilingual performance is retained through CPT, despite task-level gains and losses**.

Spanish benchmarks

benchmark	base	C1	Gain C1	C2	Gain C2	C3	Gain C3	C4	Gain C4	C5	Gain C5	C6	Gain C6	C7	Gain C7	C8
Global MMLU Lite ES (acc)	**70.00 (65.50, 74.50)**	63.75 (59.03, 68.47)	-6.25	63.50 (58.78, 68.22)	-6.50	63.75 (59.03, 68.47)	-6.25	62.75 (58.01, 67.49)	-7.25	63.25 (58.52, 67.98)	-6.75	63.75 (59.03, 68.47)	-6.25	63.50 (58.78, 68.22)	-6.50	63.25 (58.52, 67.98)
Belebele ES (acc)	**84.78 (82.43, 87.13)**	78.56 (75.88, 81.24)	-6.22	77.89 (75.18, 80.60)	-6.89	79.11 (76.45, 81.77)	-5.67	78.22 (75.52, 80.92)	-6.56	78.67 (75.99, 81.35)	-6.11	79.56 (76.92, 82.20)	-5.22	80.33 (77.73, 82.93)	-4.45	79.78 (77.15, 82.41)
MultiWikiQA ES (F1)	**79.63 (78.72, 80.54)**	70.36 (69.32, 71.40)	-9.27	68.14 (67.06, 69.22)	-11.49	70.27 (69.22, 71.32)	-9.36	69.63 (68.57, 70.69)	-10.00	71.19 (70.15, 72.23)	-8.44	71.25 (70.20, 72.30)	-8.38	70.76 (69.71, 71.81)	-8.87	70.15 (69.10, 71.20)
ARC Easy ES (acc_norm)	**89.47 (86.95, 91.99)**	86.49 (83.68, 89.30)	-2.98	87.02 (84.26, 89.78)	-2.45	88.77 (86.18, 91.36)	-0.70	87.72 (85.02, 90.42)	-1.75	82.63 (79.52, 85.74)	-6.84	88.07 (85.41, 90.73)	-1.40	87.37 (84.64, 90.10)	-2.10	87.54 (84.83, 90.25)
ARC Challenge ES (acc_norm)	**81.94 (77.57, 86.31)**	79.26 (74.66, 83.86)	-2.68	77.26 (72.50, 82.02)	-4.68	80.27 (75.75, 84.79)	-1.67	77.26 (72.50, 82.02)	-4.68	79.26 (74.66, 83.86)	-2.68	80.27 (75.75, 84.79)	-1.67	77.59 (72.86, 82.32)	-4.35	78.93 (74.30, 83.56)
PIQA ES (acc_norm)	**83.00 (75.60, 90.40)**	79.00 (70.98, 87.02)	-4.00	78.00 (69.84, 86.16)	-5.00	76.00 (67.59, 84.41)	-7.00	79.00 (70.98, 87.02)	-4.00	80.00 (72.12, 87.88)	-3.00	77.00 (68.71, 85.29)	-6.00	79.00 (70.98, 87.02)	-4.00	78.00 (69.84, 86.16)
HellaSwag ES (acc_norm)	**73.49 (72.60, 74.38)**	71.74 (70.83, 72.65)	-1.75	71.64 (70.73, 72.55)	-1.85	71.75 (70.84, 72.66)	-1.74	71.86 (70.95, 72.77)	-1.63	72.58 (71.68, 73.48)	-0.91	72.27 (71.36, 73.18)	-1.22	71.83 (70.92, 72.74)	-1.66	72.05 (71.14, 72.96)
COPA ES (acc)	82.00 (74.43, 89.57)	76.00 (67.59, 84.41)	-6.00	82.00 (74.43, 89.57)	+0.00	79.00 (70.98, 87.02)	-3.00	80.00 (72.12, 87.88)	-2.00	78.00 (69.84, 86.16)	-4.00	79.00 (70.98, 87.02)	-3.00	**83.00 (75.60, 90.40)**	+1.00	77.00 (68.71, 85.29)
XStoryCloze ES (acc)	75.25 (73.07, 77.43)	73.66 (71.44, 75.88)	-1.59	74.59 (72.39, 76.79)	-0.66	74.12 (71.91, 76.33)	-1.13	**75.31 (73.14, 77.48)**	+0.06	74.72 (72.53, 76.91)	-0.53	74.72 (72.53, 76.91)	-0.53	74.39 (72.19, 76.59)	-0.86	75.18 (73.00, 77.36)
TruthfulQA-MC1 ES (acc)	**32.07 (28.81, 35.33)**	29.15 (25.98, 32.32)	-2.92	31.94 (28.68, 35.20)	-0.13	28.26 (25.12, 31.40)	-3.81	30.80 (27.58, 34.02)	-1.27	29.28 (26.10, 32.46)	-2.79	28.52 (25.37, 31.67)	-3.55	27.63 (24.51, 30.75)	-4.44	28.52 (25.37, 31.67)
WNLI ES (acc)	43.66 (32.04, 55.28)	52.11 (40.41, 63.81)	+8.45	45.07 (33.41, 56.73)	+1.41	47.89 (36.19, 59.59)	+4.23	40.85 (29.33, 52.37)	-2.81	43.66 (32.04, 55.28)	+0.00	43.66 (32.04, 55.28)	+0.00	42.25 (30.68, 53.82)	-1.41	**59.15 (47.63, 70.67)**
MGSM ES (exact_match)	**27.20 (21.67, 32.73)**	19.60 (14.67, 24.53)	-7.60	23.60 (18.33, 28.87)	-3.60	17.60 (12.87, 22.33)	-9.60	20.40 (15.39, 25.41)	-6.80	22.00 (16.85, 27.15)	-5.20	18.40 (13.59, 23.21)	-8.80	22.40 (17.22, 27.58)	-4.80	15.20 (10.74, 19.66)
GPQA Main ES (acc)	31.03 (26.74, 35.32)	28.12 (23.95, 32.29)	-2.91	29.69 (25.45, 33.93)	-1.34	31.47 (27.16, 35.78)	+0.44	30.36 (26.10, 34.62)	-0.67	27.68 (23.53, 31.83)	-3.35	25.45 (21.41, 29.49)	-5.58	28.12 (23.95, 32.29)	-2.91	**31.92 (27.60, 36.24)**
Average	**65.65 (64.18, 67.12)**	62.14 (60.62, 63.66)	-3.51	62.33 (60.82, 63.84)	-3.32	62.17 (60.66, 63.68)	-3.48	61.86 (60.36, 63.36)	-3.79	61.76 (60.25, 63.27)	-3.89	61.69 (60.19, 63.19)	-3.96	62.17 (60.69, 63.65)	-3.48	62.82 (61.32, 64.32)

Note: In bold the highest value per row. In parentheses the 95% confidence intervals (CI), which are approximate normal intervals computed as the score or average ± 1.96 times its standard error. Gain columns show signed score-point differences from the corresponding base model.

English benchmarks

benchmark	base	C1	Gain C1	C2	Gain C2	C3	Gain C3	C4	Gain C4	C5	Gain C5	C6	Gain C6	C7	Gain C7	C8
Global MMLU Lite EN (acc)	**74.25 (69.96, 78.54)**	68.00 (63.42, 72.58)	-6.25	65.00 (60.32, 69.68)	-9.25	66.50 (61.87, 71.13)	-7.75	66.75 (62.13, 71.37)	-7.50	64.75 (60.06, 69.44)	-9.50	68.50 (63.94, 73.06)	-5.75	69.00 (64.46, 73.54)	-5.25	66.25 (61.61, 70.89)
Belebele EN (acc)	**89.78 (87.80, 91.76)**	84.33 (81.95, 86.71)	-5.45	85.22 (82.90, 87.54)	-4.56	83.33 (80.89, 85.77)	-6.45	85.56 (83.26, 87.86)	-4.22	83.67 (81.25, 86.09)	-6.11	87.44 (85.27, 89.61)	-2.34	86.33 (84.08, 88.58)	-3.45	85.22 (82.90, 87.54)
MultiWikiQA EN (F1)	**81.52 (80.64, 82.40)**	74.09 (73.10, 75.08)	-7.43	71.49 (70.45, 72.53)	-10.03	73.70 (72.68, 74.72)	-7.82	73.09 (72.06, 74.12)	-8.43	73.35 (72.33, 74.37)	-8.17	74.39 (73.38, 75.40)	-7.13	73.90 (72.89, 74.91)	-7.62	73.75 (72.73, 74.77)
ARC Easy EN (acc_norm)	**83.51 (80.46, 86.56)**	78.77 (75.41, 82.13)	-4.74	77.54 (74.11, 80.97)	-5.97	78.07 (74.67, 81.47)	-5.44	79.47 (76.15, 82.79)	-4.04	77.72 (74.30, 81.14)	-5.79	80.53 (77.28, 83.78)	-2.98	77.72 (74.30, 81.14)	-5.79	77.54 (74.11, 80.97)
ARC Challenge EN (acc_norm)	**65.22 (59.81, 70.63)**	59.87 (54.30, 65.44)	-5.35	58.86 (53.27, 64.45)	-6.36	60.54 (54.99, 66.09)	-4.68	56.19 (50.56, 61.82)	-9.03	62.54 (57.04, 68.04)	-2.68	61.20 (55.67, 66.73)	-4.02	61.20 (55.67, 66.73)	-4.02	63.55 (58.09, 69.01)
PIQA EN (acc_norm)	**83.00 (75.60, 90.40)**	79.00 (70.98, 87.02)	-4.00	80.00 (72.12, 87.88)	-3.00	82.00 (74.43, 89.57)	-1.00	80.00 (72.12, 87.88)	-3.00	82.00 (74.43, 89.57)	-1.00	81.00 (73.27, 88.73)	-2.00	79.00 (70.98, 87.02)	-4.00	80.00 (72.12, 87.88)
HellaSwag EN (acc_norm)	**81.29 (80.53, 82.05)**	80.59 (79.82, 81.36)	-0.70	80.11 (79.33, 80.89)	-1.18	80.66 (79.89, 81.43)	-0.63	79.67 (78.88, 80.46)	-1.62	80.84 (80.07, 81.61)	-0.45	80.58 (79.81, 81.35)	-0.71	80.61 (79.84, 81.38)	-0.68	80.60 (79.83, 81.37)
WNLI EN (acc)	30.99 (20.16, 41.82)	35.21 (24.02, 46.40)	+4.22	32.39 (21.43, 43.35)	+1.40	32.39 (21.43, 43.35)	+1.40	32.39 (21.43, 43.35)	+1.40	**39.44 (27.99, 50.89)**	+8.45	30.99 (20.16, 41.82)	+0.00	33.80 (22.72, 44.88)	+2.81	33.80 (22.72, 44.88)
COPA EN (acc)	88.00 (81.60, 94.40)	**89.00 (82.84, 95.16)**	+1.00	86.00 (79.16, 92.84)	-2.00	88.00 (81.60, 94.40)	+0.00	88.00 (81.60, 94.40)	+0.00	87.00 (80.38, 93.62)	-1.00	85.00 (77.97, 92.03)	-3.00	87.00 (80.38, 93.62)	-1.00	88.00 (81.60, 94.40)
XStoryCloze EN (acc)	80.61 (78.62, 82.60)	80.54 (78.54, 82.54)	-0.07	81.34 (79.37, 83.31)	+0.73	81.01 (79.03, 82.99)	+0.40	81.67 (79.72, 83.62)	+1.06	81.47 (79.51, 83.43)	+0.86	**81.93 (79.99, 83.87)**	+1.32	81.80 (79.85, 83.75)	+1.19	81.87 (79.93, 83.81)
MGSM EN (exact_match)	**30.80 (25.07, 36.53)**	20.80 (15.76, 25.84)	-10.00	18.40 (13.59, 23.21)	-12.40	14.40 (10.04, 18.76)	-16.40	14.00 (9.69, 18.31)	-16.80	30.00 (24.31, 35.69)	-0.80	21.60 (16.49, 26.71)	-9.20	26.40 (20.92, 31.88)	-4.40	20.80 (15.76, 25.84)
GPQA Main EN (acc)	32.14 (27.81, 36.47)	34.60 (30.19, 39.01)	+2.46	31.03 (26.74, 35.32)	-1.11	**35.04 (30.62, 39.46)**	+2.90	32.37 (28.03, 36.71)	+0.23	32.14 (27.81, 36.47)	+0.00	32.37 (28.03, 36.71)	+0.23	34.60 (30.19, 39.01)	+2.46	33.71 (29.33, 38.09)
TruthfulQA-MC1 EN (acc)	**28.40 (25.31, 31.49)**	24.24 (21.30, 27.18)	-4.16	25.46 (22.47, 28.45)	-2.94	23.50 (20.59, 26.41)	-4.90	24.72 (21.76, 27.68)	-3.68	26.07 (23.06, 29.08)	-2.33	24.97 (22.00, 27.94)	-3.43	24.24 (21.30, 27.18)	-4.16	24.24 (21.30, 27.18)
Average	**65.35 (63.93, 66.77)**	62.23 (60.78, 63.68)	-3.12	60.99 (59.54, 62.44)	-4.36	61.47 (60.05, 62.89)	-3.88	61.07 (59.64, 62.50)	-4.28	63.15 (61.68, 64.62)	-2.20	62.35 (60.91, 63.79)	-3.00	62.74 (61.28, 64.20)	-2.61	62.26 (60.81, 63.71)

Note: In bold the highest value per row. In parentheses the 95% confidence intervals (CI), which are approximate normal intervals computed as the score or average ± 1.96 times its standard error. Gain columns show signed score-point differences from the corresponding base model.

The language-level averages (see table below) show the same broad pattern: the Guarani average increases by `3.13`–`6.15` points over the base score, while `English` remains within `2.01` points of its base average and Spanish within `1.94` points of its base average in the worst cases. **These averages further indicate stronger Guarani performance alongside substantial retention of English and Spanish**.

Variant	Guarani Average	Guarani Gain	English Average	English Gain	Spanish Average	Spanish Gain
base	31.66	--	**59.73**	--	**59.75**	--
C1	36.19	+4.53	58.88	-0.85	58.29	-1.46
C2	35.09	+3.43	57.72	-2.01	58.34	-1.40
C3	35.58	+3.93	58.25	-1.47	58.33	-1.42
C4	35.55	+3.89	57.81	-1.92	58.01	-1.73
C5	36.07	+4.41	59.71	**-0.02**	57.94	-1.80
C6	34.78	+3.13	58.88	-0.85	57.80	-1.94
C7	37.42	+5.76	59.34	-0.39	58.33	-1.42
C8	**37.81**	**+6.15**	58.83	-0.90	58.82	**-0.93**

Note: In bold the highest value per column. Gain columns show signed score-point differences from the corresponding base model.

Data configuration finding

`C7` shows to be the most promising data configuration to carry forward to supervised fine-tuning. It has the best perplexity results, leads translation from Guarani, scores highest on Guarani `Belebele`, and comes closest to the base on Guarani `Global MMLU Lite` and `MGSM`. It also retains multilingual performance well: its English average ranks second among CPT variants, and its Spanish average is among the strongest.

> [!IMPORTANT] While no single configuration leads every evaluation, the dataset configuration C7 shows to offer the strongest balance across the Guarani capabilities and multilingual retention.

Finding the CPT method

With `C7` selected as the most promising data configuration, the next step is to compare `full CPT` with `LoRA`. The comparison uses `Gemma-3 E4B` models trained on `C7`, with LoRA ranks of `64`, `128`, `256`, and `512`.